Race and Persona in Imprecision Interpretation in the Visual World Paradigm

Recent evidence suggests that precision thresholds in numeral interpretation may be adjusted in complex ways based on aspects of speakers' social identities. In particular, comprehenders are less prone to choose a displayed time of "2:51" as fitting the utterance "It's 3:00" with nerdy-looking (more precise) vs. chill-looking (less precise) speakers when those speakers are white ([1]); yet the Nerd–Chill difference does not arise (or potentially even flips) for Black speakers ([2]). Moreover, comprehenders are overall more likely to reject the imprecise "2:51" in such cases for Black speakers than White speakers; [2] suggests that the greater precision for Black speaker utterances may reflect comprehenders' awareness that Black speakers are more exposed to social risks associated with imprecision (e.g., being accused of misrepresenting facts). [2] further proposes that Black Nerds' closer proximity to whiteness attenuates their risk effect, offsetting the persona difference found for White speakers in [1]. We use visual world eye-tracking to dissociate these two factors, and to assess when and how they enter online comprehension of imprecise statements.

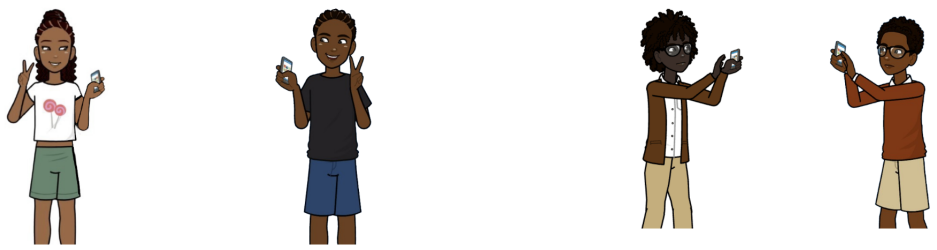
DESIGN. We use an eye-tracking task with a covered box paradigm that parallels prior behavioral work. Participants saw dialogues with one character asking a question and another responding with a numeral utterance after checking their phone. 3 factors were crossed in a 2x2x3 design:

- **Speaker Persona** (b-subj): *Nerd* vs. *Chill* (expected to be more vs. less precise, Fig. 1)
- **Speaker Race** (b-subj): *Black* vs. *White* (expected to be interpreted more vs. less precisely)
- Display/utterance **Match** (w-subj): *Match* (3:00); *MisMatch* (5:53); *Imprecise* (2:51)

Participants (n=160) saw 24 items (+24 fillers) from one of 6 lists and judged whether a visible phone or one where the numbers were not visible was the one that the speaker was looking at (Fig. 2). Covered choices in the critical *Imprecise* condition indicate a precise interpretation.

RESULTS. Descriptive statistics of response patterns align with prior work, with numerical differences in covered choice rates in opposite directions and elevated rates for Black speakers (Fig. 3). Given that participants generally first inspected the visible phone (always on left), our initial eye tracking analysis on percent looks to covered phone in the *Imprecise* condition focuses on the time window from 1000ms after display of the phones (counting from the onset of the auditory answer stimulus) until 2000ms (by when about 50% of trials have ended due to a response choice having been made); see Fig. 4. We split this time-window in half to assess earlier vs. later effects, leading to a 2$\times$2$\times$2 interaction analysis of Persona, Speaker Race, and Time window. Proportions of looks to covered picture were transformed to Elogits, and analyzed in an LMEM with random slopes for Time window by participant and Race by item. We find a 3-way interaction ($p < .05$); post-hoc comparisons find the interaction to be driven by a Persona$\times$Time Window interaction ($p < .05$) for Black speakers (with more initial looks to covered for Nerd and more later looks to covered for Chill) that is absent for White speakers ($p = .2$), where an initial pattern parallel to that for Black speakers continues in the later window.

DISCUSSION. Our results show the same pattern of responses as in [1] & [2]. The eye movement pattern are in line with [2]'s proposal that two factors play together: Persona affects a first phase of fixation patterns with more looks to the covered picture for regardless of race. Only in a later phase do the eye movements align with opposite response patterns for Black and White speakers. In addition to providing evidence for separating out these two factors, this suggests that listeners can rapidly integrate the socioindexical association between precision and the Nerd persona, consistent with prior work on automated persona processing ([3]), but need longer for the higher-order pragmatic reasoning involved in computing speakers' social risk calculations (here triggered by race) and forming expectations about the resulting behavior. This opens up novel perspectives for investigating the interplay of different social factors affecting linguistic interpretation in online processing.



(a) *Chill* pair. (b) *Nerdy* pair.

Figure 1: Persona manipulation.

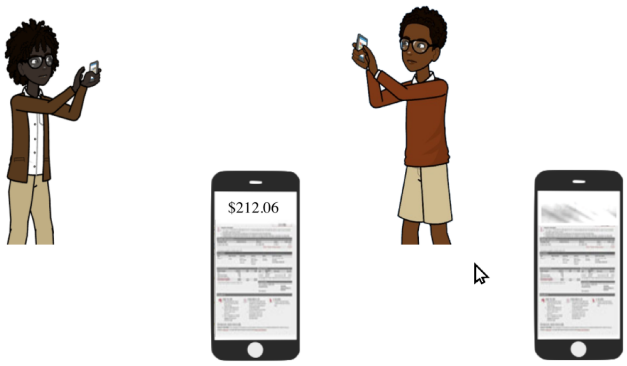


Figure 2: Example trial from the *Imprecise* condition with *Nerdy* characters. Target utterance: "It's $200."

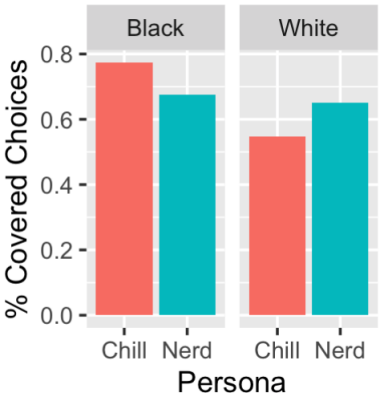


Figure 3: Overall FALSE choices by Match

Figure 4: Percent Looks to Covered by Speaker Race

References. [**1**] Beltrama & Schwarz 2024. (Im)precise personae: The effect of socio-indexical information on semantic interpretation. *Language in Society*, *54*(4), 787-814. • [**2**] Martin & Beltrama 2025. The effects of persona racialization on perceptions of pragmatic (im)precision. *NWAV 53* • [**3**] D'Onofrio 2018. Controlled and automatic perceptions of a sociolinguistic marker. *Language Variation and Change*, *30*(2), 261-285. •